

Introduction to Nanotechnology

Unit- 1 Introduction to Nanotechnology

What is Nano?

The word 'nano' is from the Greek word 'Nanos' meaning Dwarf. It is a prefix used to describe "one billionth" of something.

A nanometre (nm) is a billionth of a metre, or 10^{-9} , a millionth of a millimetre.

1 nanometre is about 8 times the radius of an atom and 100 times smaller than a bacterial cell.

Nanoscale: having one or more dimensions of the order of 100 nm or less.

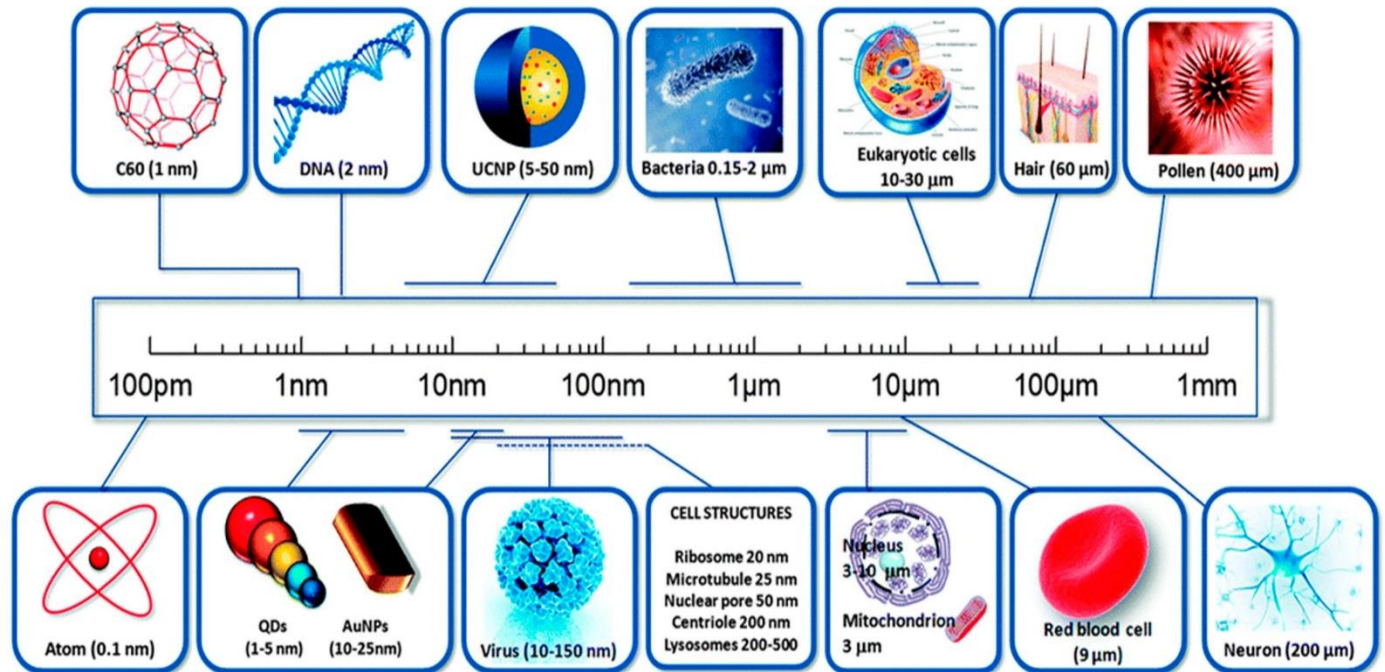
Physical dimensions of world

At first, it can be hard to comprehend the nanoscale because it is so much smaller than our everyday experience. While we know intuitively that a dime is smaller than a basketball, and even that a red blood cell (which can be observed in the light microscope) is smaller than a marble, we have no experience with objects that are billionths (10^{-9}) of a meter (1 nanometer or nm) in length. When was the last time you put your hands around a strand of DNA (2.5 nm) or measured the diameter of a flu virus (100 nm)? Here are a few comparisons to help understand how small a nanometer is:

- An average human hair is about 60,000 -100,000 nanometers wide
- Your fingernail grows a nanometer every second

- A sheet of paper is about 100,000 nanometers thick
- In one inch there are 25,400,000 nanometers

To explore the size of the nanoscale in relation to other scales there are some great interactive sites such as [Scale of the Universe](#), [Powers of Ten](#), and [Nanosize Me](#).



Nanotechnology is often divided into two parts. Nanoscience & Nanotechnology.

Nanoscience: Nanoscience: the study of phenomena and manipulation of materials at atomic, molecular and macromolecular scales, where properties differ significantly from those at a larger scale.

Nanotechnology: the design, characterization, production and application of structures, devices and systems by controlling shape and size at the nanoscale.

Nanomaterial: material with one or more external dimensions, or an internal structure, which could exhibit novel characteristics compared to the same material without nanoscale features.

According to Dimension Nano materials are classified in three types(0D,1D,2D) :Quantum dot(0D), quantum wire(1D) & quantum well(2D)

Quantum dots: The extreme case of this process of size reduction in which all three dimensions reach the low nanometer range is called a quantum dot.

Quantum wire :If two dimensions are so reduced and one remains large, the resulting structure is referred to as a quantum wire.

Quantum wells: If one dimension is reduced to the nano range while the other two dimensions remain large. then we obtain a structure known as a quantum well.

Classification of Techniques for Synthesis of Nanomaterial: Top -Down and Bottom up Approach

here are two general approaches for the synthesis of nanomaterials as shown in Figure.

(a) Top-down approach

The top-down approach involves the breaking down of the bulk material into nanosized structures or particles. So basically in a top-down approach, a large-scale object is progressively reduced in dimensions up to nanoscale.

Top-down approaches are inherently simpler and depend either on removal or division of bulk material or on miniaturization of bulk

fabrication processes to produce the desired structure with appropriate properties.

This method is very time consuming and relatively costly, it produce much waste and the biggest problem with the top-down approach is the imperfection of surface structure.

For example, nanowires made by lithography are not smooth and may contain a lot of impurities and structural defects on their surface. Examples of such techniques are high-energy wet ball milling, electron beam lithography, atomic force manipulation, gas-phase condensation, aerosol spray, etc.

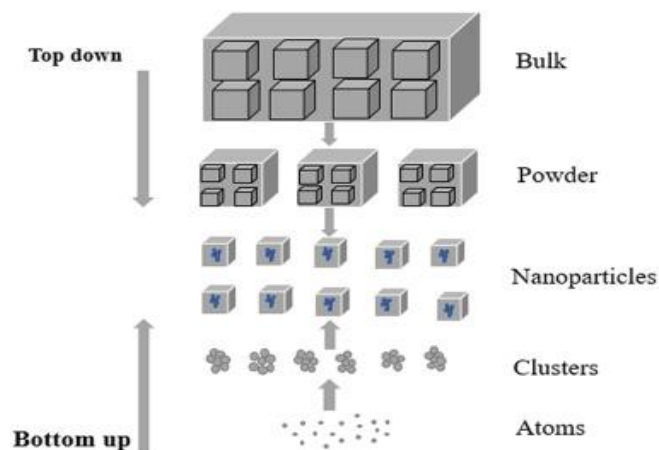


Figure 1 : Schematic representation of ‘top-down’ and 'bottom-up' approaches for the synthesis of nanoscale materials.

(b) Bottom-up approach

- The alternative approach, which has the potential of creating less waste and hence the more economical, is the ‘bottom-up’.
- Bottom-up approach refers to the build-up of material from the bottom: atom-by-atom, molecule-by-molecule, or cluster-by-cluster

- It is a chemical process in which the different materials & devices are constructed from molecular components on their own which do not require any external agent to assemble them.
- They chemically assemble themselves by recognizing the molecules by their type and this start by collecting and combining the atom and molecule to build complex structure.
- Organometallic chemical route, reverse-micelle route, sol-gel synthesis, colloidal precipitation, hydrothermal synthesis, template assisted sol-gel, electemplate-assisted, are some of the well-known bottom-up techniques reported for the preparation of luminescent nanoparticles.

Unit -2 : Types of Nanostructures

Fundamental carbon based nanostructures.

Carbon materials are found in variety forms such as graphene, graphite, diamond, carbon fibers, fullerenes, and carbon nanotubes.

Carbon is the only material in the periodic table that has isomers from 0 dimensions (0D) to 3 dimensions (3D), as is shown in table bellow.

Carbon Hybridization

Structure	0-D	1-D	2-D	3-D
Isomer	C60 fullerene	nanotube	Graphite fiber	diamond
Hybridization	sp ²	sp ²	sp ²	sp ³
Bond Length [Å]	1.40 (C=C) 1.46 (C-C)	1.44 (C=C)	1.42 (C=C)	1.54 (C-C)
Electronic Properties	Semiconductor E _g =1.9 eV	Metal or semiconductor	semimetal	Insulator E _g = 5.47eV

Carbon has been studied and used for centuries, and carbon science was long thought to be a mature field. Diamond (3D) and graphite (2D)

are considered as two natural crystalline forms of pure carbon. So when a whole new class of carbon materials—the **fullerenes**, appeared in the last few decades,

A **fullerene** is an **allotrope of carbon** whose molecules consist of carbon atoms connected by single and double bonds so as to form a closed or partially closed mesh, with fused rings of five to seven atoms. The molecules may be hollow **spheres**, **ellipsoids**, **tubes**, or other shapes. When the carbon molecules are arranged in a cylindrical form they usually form a tube-like structure known as carbon nanotubes.

Fullerenes with a closed mesh topology are informally denoted by their **empirical formula** C_n , often written C_n , where n is the number of carbon atoms. However, for some values of n there may be more than one **isomer**.

there is an infinite number of fullerenes that can exist. Fullerenes can be C_{60} , C_{70} , C_{80} , and C_{90} existing in various forms.

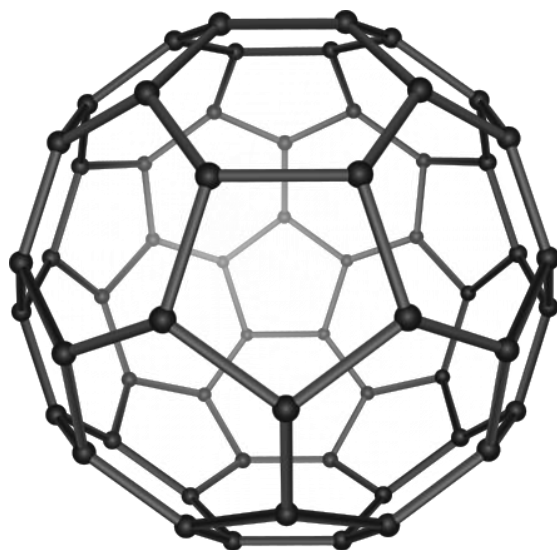
The family is named after **buckminsterfullerene** (C_{60}), the most famous member, which in turn is named after **Buckminster Fuller**.

C_{60} is the third major form of pure carbon; graphite and diamond are the other two."

Structure of Fullerene C_{60} (Bucky ball)

Buckminster-fullerene is the smallest fullerene molecule. Molecules made up of 60 carbon atoms arranged in a series of interlocking hexagons and pentagons, forming a structure that looks similar to a soccer ball. C_{60} is actually a "truncated icosahedron", consisting of 12 pentagons and 20 hexagons and no two pentagons will share an edge.

The buckyball, being the roundest of round molecules, is also quite resistant to high speed collisions. In fact, the buckyball can withstand slamming into a stainless steel plate at 15,000 mph, merely bouncing back, unharmed. When compressed to 70 percent of its original size, the buckyball becomes more than twice as hard as its cousin, diamond."



Model of the C₆₀ fullerene (buckminsterfullerene)

Uses of Fullerene

- Buckminsterfullerene is used in drug delivery systems, in lubricants and as a catalyst.
- It is also used in several biomedical applications including the design of high-performance MRI contrast agents, X-ray imaging contrast agents, photodynamic therapy and drug and gene delivery
- It is also used as a conductor.
- Some types of fullerene can be used as an absorbent for gases.
- It is used in making cosmetic products.

- C60 based films are used for photovoltaic applications.
- Fullerenes are used in making carbon nanotubes-based fabrics and fibres. Etc

Carbon Nanotubes

carbon nanotube is a hollow tube made up of carbon of nanoscale diameter. In short, it is represented as CNTs. Carbon nanotubes are also called buckytubes. Nanotubes are formed by folding or rolling two-dimensional graphite into a cylindrical shape structure.

Nanotubes are hollow from inside. The diameter of the nanotube is around 1-3 nanometers. The length of the carbon nanotube is much higher than its diameter. Nanotube length generally goes to a few micrometers. In short, we can say that carbon nan (CNT) is a folded form of the two-dimensional graphene sheet. CNT (carbon nanotubes) exhibit extraordinary mechanical properties.

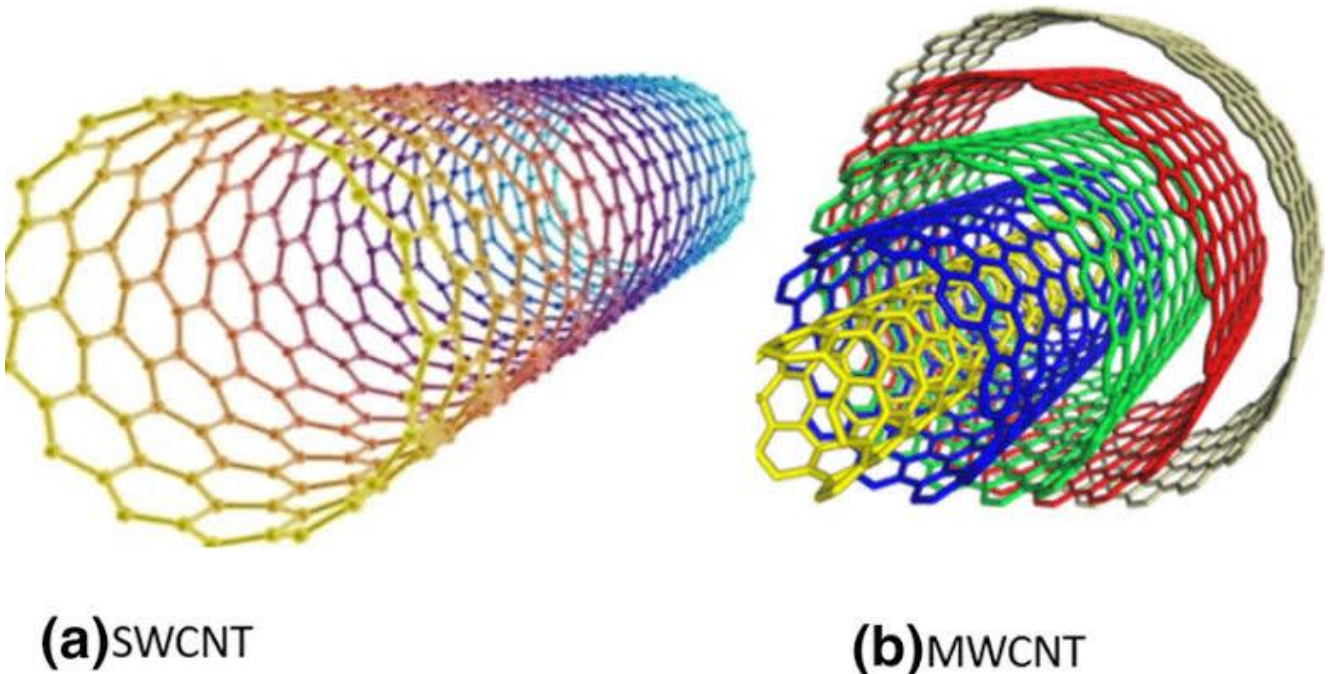
Types of Carbon Nanotubes

1. Single-walled Carbon nanotubes (SWCNT)
2. Multi-walled Carbon nanotubes (MWCNT)

Single-walled Carbon Nanotubes- it is represented as SWCNT. The Single-walled Carbon nanotubes exist in a 1-d structure. Some examples of Single-walled CNT are armchair and zig-zag Single-walled Carbon nanotubes. The diameter of Single-walled Carbon

nanotubes is 2nm. Its length is around 2 micrometers. They exist in a one-dimensional structure. Therefore, it is also known as a nanowire.

Multi-walled Carbon Nanotubes- It is represented as MWCNT. It is composed of several nested carbon nanotubes. This type of nanotubes has two diameters, outer diameter and another one is inner diameter. The outer diameter of Multi-walled Carbon nanotubes is around 2-20 nanometres. The inner diameter of Multi-walled Carbon nanotubes is 1-3 nm. The length of Multi-walled Carbon nanotubes is around 5-6 micrometers. An example of Multi-walled Carbon nanotubes is chiral Multi-walled Carbon nanotubes.



Application of carbon nanotubes

- Breast cancer tumor destruction:nanotubes are used to destroy breast cancer tumors. They play with an antibody.

The antibody along with nanotubes is attracted to the proteins by cancer cells in the body and nanotubes absorb the laser beam killing the bacteria of the tumor.

- Windmill blades: carbon nanotubes are also used in the windmill blades because of their low weight . It increases the efficiency of the windmill and helps to produce more electricity at a faster rate.
- Filtration: carbon nanotubes can be used to separate particles of size greater than the diameter of carbon nanotubes during filtration through them. They can also be used to trap smaller sized ions from a solution.
- Carbon nanotubes as Nano cylinders: gas like H_2 , for energy, battery for vehicles can be safely stored inside the carbon nanotubes and the problem of H_2 storage hazards can be solved.
- Carbon nanotubes have also been shown to absorb infrared light and may have applications in the IR optics industry.
- Aircraft stress reduction: nanotubes are also used in space and aircraft to reduce the weight and stress of the various components working together